DICE

Decentralized Infrastructure for Cryptographic Entropy

Hardware-Backed Verifiable Randomness
for Solana

Solana | Hardware Entropy | DePIN | VRF

Physical Devices → Generate Entropy → Verify on Solana → dApps Get Randomness

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Product Concept

ONE-LINER

The Cloudflare of on-chain randomness — powered by physical hardware devices.

- DICE is a distributed randomness network powered by physical hardware nodes.
- Small internet-connected ESP32-S3 devices generate hardware-backed entropy.
- Multiple independent nodes operate across different locations and networks.
- Applications request randomness through a Solana smart contract.
- Results are cryptographically verifiable and delivered in seconds.

Use Cases

- On-Chain Games
- Lotteries
- NFT Mints
- Prediction Markets
- Governance

No cloud servers — physical devices generate real-world entropy

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Target Users

Solana dApp Developers

- Games & lotteries needing fair RNG
- NFT mint platforms
- Integrate via smart contract
- Replacing Switchboard / Chainlink VRF

DeFi Protocol Teams

- Prediction markets
- Governance selection
- Fair token launches
- Need tamper-proof, fast finality

Node Operators & HW Enthusiasts

- Already run validators or IoT
- Plug-and-play device
- Earn per entropy contribution
- DePIN infrastructure believers

Find them on Discord · Telegram · GitHub · X

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Problems

Real problems observed in the ecosystem — not imagined, proven.

01

Blockchains are deterministic

Smart contracts can't generate randomness. Every validator must compute the same result. True randomness cannot originate from the chain itself.

02

Current VRF solutions are centralized

Oracle-based randomness relies on server nodes run by a few operators. Single points of failure destroy the trust model that decentralization promises.

03

No hardware-level entropy layer exists

There is no decentralized network providing physical entropy. Software-only RNGs are predictable, manipulable, and lack real-world unpredictability.

04

High cost and limited access

Randomness requests on other chains are expensive and slow. Smaller dApp teams can't justify the cost or latency of existing solutions for every request.

04

Solutions

A structural change — not just another feature, product, or oracle.

solves: can't generate randomness

External Hardware Entropy Layer

ESP32-S3 nodes generate entropy from hardware RNG, clock jitter, and environmental noise — real physical randomness from the real world.

solves: centralized VRFs

Distributed Device Network

Independent hardware nodes across locations. Commit-reveal protocol ensures integrity — even if most nodes are malicious, one honest node keeps it secure.

solves: no hardware entropy

Cryptographic Proof Per Device

Each device signs its entropy with a private key. Every single contribution is verifiable back to its physical source device.

solves: high cost

Solana-Native, Sub-Cent Delivery

Built on Solana smart contracts. Sub-cent transaction fees, 2-5 second finality, accessible for any team size.

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Why Solana

Low Fees

Sub-cent per txn.
Thousands daily.

Fast Finality

2-5 second delivery.
~400ms slot times.

High TPS

65,000+ transactions
per second.

Ecosystem

Games, NFTs, DeFi
already need RNG.

What DICE Contributes Back to Solana

New Infrastructure Primitive

Fills a missing randomness layer the ecosystem lacks.

DePIN Growth

Hardware operators join Solana — expanding beyond storage.

Developer Tooling

Open SDK lowers the barrier for any developer.

On-Chain Economic Loop

Apps pay, nodes earn — self-sustaining protocol.

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How It Works

Six steps from request to verified randomness — under 5 seconds.

1 Request

dApp calls requestRandomness()

2 Select

Coordinator picks 15 of 50 nodes

3 Generate

Each node creates hardware entropy

4 Commit

Nodes submit hash(entropy)

5 Reveal

Nodes reveal entropy
Network verifies

6 Deliver

Final randomness written on-chain

Commit-Reveal Protocol — Why It's Tamper-Proof

1. Commit

Each node hashes entropy before sharing anything

`hash(entropy) → commit`

2. Reveal

Nodes reveal raw entropy
Network verifies hashes

`hash(revealed) == commit?`

3. Combine

All entropy combined into one final random value

`hash(e1 + e2 + e3 ...)`

Even if most nodes are malicious, one honest node keeps the result secure

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System Architecture

Application Layer

dApps · Games · Lotteries · DeFi

`Game` `Lottery` `NFT Mint` `DeFi`

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Smart Contract

Solana Program · Requests · Results · Rewards

`requestRandom()` `verifyResult()` `distributeReward()`

↓

Coordinator

Node Selection · Commit-Reveal · Aggregation

`Select Nodes` `Manage Protocol` `Aggregate`

↓

Hardware Layer

ESP32-S3 Physical Nodes · Distributed Globally

`Node 1` `Node 2` `Node 3` `Node 4` `Node 5`

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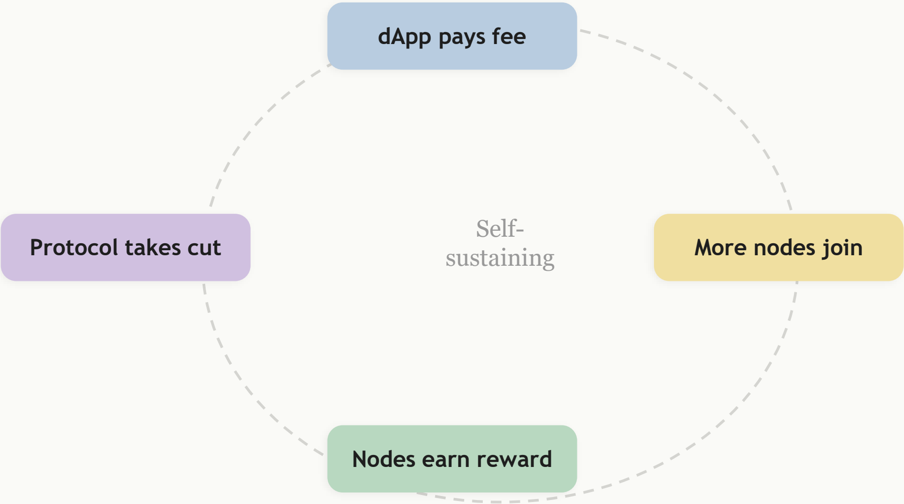
Economics & Roadmap

Economic Loop



Fee Split

70% → Node Operators | 30% Protocol

Roadmap

Phase 1 — **Closed Alpha**

Devices to trusted devs
Test protocol stability

Phase 2 — **Developer Preview**

Launch Solana SDK
First dApp integrations

Phase 3 — **Mainnet Launch**

Public network live
Reward distribution

Phase 4 — **Scale & Expand**

Global deployment
Cross-chain support

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In One Look

DICE	— Decentralized Infrastructure for Cryptographic Entropy.
Problem	— Blockchains are deterministic. Current VRFs are centralized.
Solution	— ESP32-S3 devices generate real entropy with commit-reveal protocol.
Security	— Even if most nodes are malicious, one honest node keeps it secure.
Why Solana	— Sub-cent fees. 2-5s finality. Massive ecosystem needing randomness.
For Developers	— Integrate via Solana smart contract. Get verifiable results.
For Operators	— Plug in device. Connect WiFi. Earn rewards for entropy.
Economics	— Apps pay per request. Nodes earn per contribution.
Vision	— A global physical randomness layer for the decentralized world.

Real entropy · Real devices · Provably fair randomness